

HR Analytics
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Week: 2
Lec 11: Data Visualization Excel - 1

Good morning. Welcome back to this Excel session. Today we will learn how to visualize the data through Excel. If you remember in a last session we had discussed what are the things that you need to prepare. So I hope you would have prepared the list of variables which is required. It is not necessary, you need to prepare the list of variables for the practice related to the applicant analysis only.

I have prepared this list of variables for the practice purpose. This is the purely dummy data. I have not collected the data from any organization. Just to make you understand, I have prepared a list of these variables and prepared this data set.

Prepare this data sheet. This data sheet is a purely dummy data. I have not collected this data from any organization, right. So let us learn about the Excel, how to use for the data visualization, okay. So as we discuss, we can prepare the list of variables, right, in the first session.

So here I am going to open this Excel and show you what are the variable list that I have prepared. So, here you can see first variable that is there that is the serial number. So, in your case if you are downloading this data it may be in application number right, it would may be application number. So this number is important because if you have to make a headcount of how many people have applied, there is no duplicate data is there. So for that, we need anyone a unique number where there is no duplicity.

So, so that we can make a headcount, how many people are there or who have applied? So, in a small number, like 100 people, 200 people have applied, and then you can manage this data easily in Excel. But when 20,000 people have applied, 25,000 people have applied, 1 lakh, 20, 15,000, Then big data is there, and in that case, it becomes very difficult to handle the data in Excel. So for that you, as I already said, we can use Power BI, Tableau, R and Studio, these are the software that we can use. The second variable that I have taken, the gender, so here you can see male and female. Third variable that I have taken that is the department, so for which department you have applied.

The fourth variable that I have taken is the interview date, on which particular date your interview is scheduled. The fifth variable that I have taken is the city, so from which city you are coming for the interview. The fifth variable that I have taken is the status of the interview, the status of the interview means whether somebody is coming for an interview or not, so that status also needs to be understood. The next variable source of the application, so from which source the application has come, whether it has come online, naukri.com, indeed.

com or it has come through LinkedIn, or it has come through the organization's website. So, what is the source of the application that also you need to understand. And then work experience what is how many years of work experience that somebody is having whoever is applying to your organization, right and next thing is the education level so these are the variables that i have developed just to make you understand to analyze the applicant profile right so that you can plan Your interview is better, right? Whatever preparation that you need to do for the interviews, interview scheduling, place, right, travelling, lodging, right, these are the arrangements that you need to make for the participants, or you have to take a call whether the interview should be online or offline, right? So, such kind of decisions that you need to take before conducting the interview inside the organisation. So, these are the variables that I have generated the data. If you want to go little bit more in-depth, then you can develop some other variables also like age, expected salary, family size, when somebody is coming for interview that person is needed accommodation or not or mode of travel.

So, such kind of why somebody is coming through air, train, bus or within the city through taxi. So, how that person is going to travel for the interview. So, these are the things may be important I am not saying only these are the variables which are important. So, how you have to select the variables? Simple thing that I am saying, only one way I am suggesting you, what questions your manager is asking you. Just think about that.

Think about that what questions your manager is asking. If your manager is asking how many applications we have received. I hope this question your manager might be asking before the interview. How many people have applied? How many are male? How many are female? What is this average age of these applicants? Candidates in a department, for which department they have applied, from which city they are coming, on which date you have scheduled the interview, on particular date how many employees are giving, candidates are supposed to give an interview, what is their family size, right? So, these are the personal information as well as related to the organization-related information that may be very very useful for you in order to make decisions about your day-to-day work. So, that is how you can prepare the list of variables first thing.

So, that is what you need to do. If you will not do then you will not be able to understand the problem very well. So, first step that I suggest to all participants whoever is attending this course, first make a list of this variable. So, if you remember I had suggested the four steps in the first session before this session I had suggested the four steps. So first step is to make a list of variable whatever problem that you are having or whatever thing that you would like to analyze in depth.

So it could be related to the recruitment. So application analysis is related to the recruitment. In the similar way you can take any selection problem, you can take any learning problem, you can take any development problem and or you can take any compensation problem or you can take any performance related problem like basic question who is the best candidate, who should be promoted, who should not be promoted. So, likewise you can take any function and based on your experience you can develop a list of these variables and after preparing the list of the variables you do the data preparation. Data preparation means you need to collect the data from various sources.

So it might be possible some of the data that I have presented here may be received from your website, or you may receive it because your website has an HRIS-like human resource management information system. So from there you can receive this information. Some of the information may be possible your HMIS is not having, HR information system is not having, it might be possible. So, in that case what you can do, you can think where that information is available and then you can collect the data from that particular source and then you can finalize your excel sheet. So, that is how you can complete the excel sheet.

So, if you remember, so this is the second step is done. The list of variables is ready, and related to that data, which is also there in Excel. So now the question comes: which data do you have to collect? So, whatever variable that you have identified is related to each function, is the data you need to collect. So now two steps are completed. Now let us come to the third step.

So, the third step is, as I was saying in the previous session, a variable that you are having is gender. So, how you would like to visualize this gender? That is what you need to, that is what you need to think about. Gender: how are you going to visualize this one through a pie chart, through a bar diagram, or through a table? All three options are available to you. It is not necessary you will visualize this gender only through pie chart or only through bar graph or only through table. You can select any option that suits your report.

whichever option you feel that you are getting the answer of your question right. So, let us assume now this table is ready and you have learnt the bar graph pie chart histogram right. Now what do you have to do with this? So first thing that you need to go do you need to go to this insert. So in insert if you will see this option is there pivot table. So here you can see two options are there one is a pivot table another option is there recommended pivot tables.

So first you can click on this recommended pivot table that you can click on. So this is the is coming so you can select the range for the recommended one second. So, first you can what you can do you can select the complete right. if you are not selecting also then also recommended fiber table automatically shows. So, here you can see these are the tables which this Excel has shown without performing anything.

Now you have to scroll it down, and then you have to check which one is suitable for you and which one is not. So some of the formats that you can see, see this data you have selected and what are the pieboard tables which are suggested. So first thing that you can see like city-wise, like so from Bangalore city how many people are coming? So you can see 2 people are coming. From Chennai how many are coming? 260 are coming. How many are coming from Delhi? 10 people are coming from Delhi.

Hyderabad, 37. And Kolkata, 56. And Mumbai, 151. And total is 516. So now you have an option. you can have in count format, you can convert into percentage format also.

So, you can convert to percentage format, you can convert, and you can see it in a number format also. So, it is up to you in which format you want to keep it. So, that is what you can see some of the tables which are recommended here. So now, if your manager is asking, if you have this table and your manager is asking how many people are coming from Bangalore, you can, immediately you can, tell this is the breakup of people who are coming for the interview, right? The same thing that you can present in a bar graph also, right? So, you can also do a bar graph so that I can tell you when I discuss When I will discuss the pivot chart, at that time I will tell you how to show through the bar graph. But now we are discussing pivot tables.

So as I was saying, what you will write on X axis, what you will write on Y, right? and what will be the title, how you want this breakup, everything that you, if you have thought early, if it is matching with the recommended one, you can go with that. If not, you want to make some changes, that also you can do it, right. So, here you can see department wise, right. So, department-wise, you can see in the same way, for HR, you can see 118 people are coming for operations highest number of people who have applied

for operations that is 140 and for services you can see that is 57 right so that is the number that you can see here Next table that you can see here that is the LinkedIn right, what are the sources and number of count is there that is what you can see right and here you can see this sum of work experience also and sum of SN also. So, this sum of work experience you may use to calculate the average also, right, average work experience candidate who are coming from Chennai.

So, that information may be useful for you how it can be useful if if you have developed some eligibility criteria or a screening criteria right and based on that by applying that you can identify how many people will come will be qualified for the position from this particular city. So, if you have this average experience, right. So, that is what you can see here. So, here you can see the sum, but the sum of the serial number. I hope you can understand there is no use of this number.

This is simple serial number if it is a sum it is not going to be the useful. In the same way you can see this sum of interview date that is also not useful data, right. So, that is how you can see sum of serial number again. So, again you can see These tables that the recommended tables, so these are the tables that you have not created. First you need to understand these tables you have not created.

These tables are suggested by the pivot table. So whichever table that you feel is giving you more relevant information, you can use that. If not then you can select the second option that is the pivot table. So, pivot table you can select. So, the moment this data is selected now option is coming new worksheet.

So, the moment you will click on this new worksheet. So, I hope you would have understood the steps, first step is the insert. So, in insert you will get two option, one is a pivot table, another one is a recommended table. So, I hope you would have seen how this recommended table, some of the relevant data that you will get it into it and some may not be useful for you, but that excel has recommended these tables which may be useful. So, whichever table that you feel is useful for you, you can go ahead with that right.

If not then you can go to this pivot table and that is how you can select and you can select new worksheet right. So, the moment you will see here new you will click on that you will get new worksheet. So, here you can see, These all variables are there which we had developed and these all variables are there in the excel sheet also. Below to these variables you can see four options are there, filters, column and rows and values right. So, here you can, I hope all of you understand what is column and what is rows in the Excel.

So, this is the row, right and this is the column right. So, this is the row and this is the column. So, why it is important to understand? Because which variable that you have to put in the row and which one you have to put into the column right. So, that is what you can understand and then you accordingly you can decide right.

So, sorry my excel got off. So, just now, so we were here, these are the variables that we had. So, we had gone to the insert, we had a pivot table from here. So, first let me select the data again. So, I have selected this data. I have selected this data, now you can go to this pivot table from here, new worksheet you have created this one.

So till here we had come, so now you can see this column and these rows. So I hope you understand what is row, what is column. So values, so which values that we want to put into the, in the columns or in a row. So first one that I have taken is the serial number. So, this is a sum of serial number is coming but there is we already know there is no use of this sum of serial number.

So, what we can do? We can click here and then we can go to this value field setting. So, here you we can see for which variable which type of analysis is required. For example, sum is required. Then we will sum it.

If count is required then we will count it. So in our case count is more meaningful than the sum and average. So whatever formula is required you can select from here and then you can click OK. So the moment you have clicked here OK. Now I want to know how many mail have applied. Here, you will see the moment I have selected the gender; here, you can see 245 female candidates have applied and 271 males have applied.

If I want to know on which particular date, how many people are going to give the interview. So, I have put a interview. So, these are the three, four dates that I have selected. First of any month that you can take, because these dates I have taken randomly, there is no logic behind it, but randomly dates I have selected. So, first one that you can see is a row 1, so in a row first one that you can see number 1, number 3, date, these are the date, first date, third date, 27 and 30th, so any month 4 dates that you can see.

Now you can see the number of application number of candidate who are going to appear in the interview. So total are 556 and these are according to date-wise. So you can give a answer to your manager or if you are preparing a report right so this table that you can make it and you can prepare in a report you in a presentation also PowerPoint presentation also. So if you feel the table is giving you good visualization you go ahead with the table. If not table then think about which other option like pie chart, bar graph, any histogram whichever thing that you believe is going to give you the better

visualization you can select that one.

But currently we are discussing about the table so I will make the all tables only. So, let us assume that your manager is asking what is the source of application. So, source of application like from where people have applied. So, 90 people that you can say applied through the website, 132 through newspaper, 57 through Naukri and 86 through LinkedIn and remaining through the indeed.

So, these are the just sources that I have selected. So, whichever that you feel right. Next thing status of interview. How many are going to appear in interview and how many are not? So, here you can see 182 person are coming for the interview and 334 people are not coming for the interview. So, what thing that you can predict? You can think about the employer brand. If your brand is so strong, then more and more people will appear in an interview.

They would like to work for your company. So, here you can see they have applied, but people are not coming for the interview. I hope this kind of analysis may be useful to understand how strong your brand is. If not coming people are keep on reducing day by day and coming people are keep on increasing, So, that indicates that your employer brand is very strong. That is the one thing. Second, every day as a manager you must be facing difficulty how many interview call you should give for one position.

So, this is the biggest dilemma that all manager might be facing. So, if you have done the trend analysis for this 4 year, 5 year, what is the trend? If you have a 5, if you are releasing 10 interview letters and how many are coming? If 10 out of 10 are coming, then you can release only 10. If out of 10 only 5 are coming and you want 1 is to 2 ratio. right, then you can identify how many people interview call that you need to give, right. So, in the same way you can see these are variables, you can keep asking the questions to yourself, right and in the same way if you want to know what is the education qualification of your candidate. So, here you can see how many are undergraduate, how many are postgraduate, right.

In the same way you can take one more important variable as a So, one as a count that I have taken, now I want to take this work experience, but sum of work experience is not good. So, if you remember what I had suggested, click here, go to the value field sheet setting and in term of work experience, average work experience may be useful. So, let us say like this is the average work experience. So, here I want to know whoever have completed the undergraduate what is the average work experience? 9.

68 years, right, Ph.D. So, here you can see 3.32 and PG 4.37. So, in the same way that

you can see what is the work experience now you can give the answer for your questions like undergraduate what is the work experience right and the same way now you can see source wise so from where most experienced people are applying website right Most experienced people are applying through the website so that is how you can see this table has come right now you can go to the source of status of interview coming or not coming. So, people who are not coming they are having less work experience and people who are coming they are having more work experience. So, such kind of even city wise also that you can calculate. So, from which city are having the highest level of work experience, Hyderabad. City wise, interview date wise that is interview date wise also you can here.

So, people who are coming the sum of interview date is not required right. So, if you want you can remove it there is no need. So, that is what I am trying to make you understand. So, that is how you can make multiple tables through this pivot table and you can answer the questions.

And how you can do these all things? Only one thing you have to practice. Practice for what? First you need to understand what, you need to make a list of the variables, you need to understand the importance of those variables, those numbers and then you have to convert into the meaningful tables and after making these tables you can interpret these tables and You can make wonderful, you can make these tables, you can take some insights and finally you can take actions in order to manage the things. So this is all about the pivot tables. In the next session we will talk, we will discuss about the pivot chart. I hope you would have learned about data visualization. In Excel right and how to use these pivot tables and pivot chart for the data visualization if so that you can answer day to day questions which are being asked by your manager as well as you can make good presentation good reports to make a presentation in front of or to submit the report to your manager on daily basis or monthly basis, quarterly basis or annually basis. Thank you.